

Osteoporosis is a common condition that is characterized by a weakness of bones, due to increased bone resorption at the cellular level, leading to an increase in the risk of fractures[1]. Osteoporosis occurs in part due to the degradation of collagen within bone cells facilitated by the Cathepsin K (CaK) protease [2]. The following proposal outlines a possible clinical phase for odanacatib, a CaK inhibitor, as osteoporosis medication [2]. Guidelines and parameters in this outline are obtained from similar parameters on past clinical trials.

To measure odanacatib efficacy, pre- and post-treatment bone mineral density will be compared, with an increase indicating efficacy [2-5]. The inclusion criteria are any post-menopausal woman that can give informed consent and have been diagnosed with osteoporosis [3-5]. The exclusion criteria are the usage of any bone homeostasis-altering drugs, any use of osteoporosis medication, any recent major bone injury, or any immunocompromised participants [2-5]. All odanacatib will be administered orally [6].

Phase I: A 1-year treatment duration, with approximately ~80 participants, designed to be placebo-controlled, dose-escalating, randomized and double-blind [7-9]. The daily dosage will increase from 5mg to 50mg [6]. Phase I participants will be frequently monitored for any adverse events.

Phase II: A 3-year treatment duration followed by 2 additional years of data collection, with approximately ~100 participants [10-11]. The design will be identical to phase I [7-11]. Participants will be more specifically observed for long-term changes, and to determine an optimal dose, using the same dose range as Phase I [7-11].

Phase III: A 2-year treatment duration with approximately 300 participants [12-13]. This phase will be randomized and double blind, however the dosage will be the optimized dosage from Phase II [12-13]. Participants will mainly be observed for the efficacy of CaK, in comparison to pre-existing medication.

Phase IV: A 5-year treatment duration with approximately 200 participants [14-15]. The design will follow Phase III's, with a modification to the dosage if needed [12-15]. Participants in this phase will be intermittently observed for long-term effects.

Total duration: 13 years.

A major ethical concern is the possibility of an adverse effect, as the inclusion criteria necessitate an already-vulnerable demographic. Although this cannot be avoided, ensuring that healthcare access is offered to all participants reporting adverse effects and

informing patients explicitly of possible adverse effects on the consent form will mitigate bad outcomes.

“Double-blind” in a clinical trial denotes that neither the participants nor the researchers know which treatment is allotted to who. Consider that participants may dismiss certain symptoms or adverse effects if they knew with certainty whether they received the placebo. Or that researchers may have expectations about the treatment group and then ‘fit’ the data to support these expectations. Double-blind designs eliminate these possible sources of bias.

“Randomized” in a clinical trial context indicates that participants are randomly assigned into the experimental and control groups. This ensures that observed changes are due to the therapeutic being studied, as otherwise the change may be a result of the pre-existing characteristic(s) used to sort participants.

References:

1. Osteoporosis [Internet]. Mayo Foundation for Medical Education and Research; 2025 [cited 2025 Oct 24]. Available from: <https://www.mayoclinic.org/diseases-conditions/osteoporosis/symptoms-causes/syc-20351968>
2. Bone HG, McClung MR, Roux C, Recker RR, Eisman JA, Verbruggen N, Hustad CM, DaSilva C, Santora AC, Ince BA. Odanacatib, a cathepsin-K inhibitor for osteoporosis: a two-year study in postmenopausal women with low bone density. *J Bone Miner Res*. 2010 May;25(5):937-47. doi: 10.1359/jbmr.091035. PMID: 19874198.
3. [Study Details | NCT05058976 | Romosozumab Use to Build Skeletal Integrity | ClinicalTrials.gov](#)
4. [Study Details | NCT02130973 | The Treatment of Osteoporosis Using a Combination of Teriparatide \(Forteo\) and Denosumab | ClinicalTrials.gov](#)
5. [Study Details | NCT05152381 | Safety of Cultured Allogeneic Adult Umbilical Cord Derived Mesenchymal Stem Cell Intravenous Infusion for Osteoporosis | ClinicalTrials.gov](#)
6. McClung MR, O'Donoghue ML, Papapoulos SE, Bone H, Langdahl B, Saag KG, Reid IR, Kiel DP, Cavallari I, Bonaca MP, Wiviott SD, de Villiers T, Ling X, Lippuner K, Nakamura T, Reginster JY, Rodriguez-Portales JA, Roux C, Zanchetta J, Zerbini CAF, Park JG, Im K, Cange A, Grip LT, Heyden N, DaSilva C, Cohn D, Massaad R, Scott BB, Verbruggen N, Gurner D, Miller DL, Blair ML, Polis AB, Stoch SA, Santora A, Lombardi A, Leung AT, Kaufman KD, Sabatine MS; LOFT Investigators. Odanacatib for the treatment of postmenopausal osteoporosis: results of the LOFT multicentre, randomised, double-blind, placebo-controlled trial and LOFT

Extension study. Lancet Diabetes Endocrinol. 2019 Dec;7(12):899-911. doi: 10.1016/S2213-8587(19)30346-8. Epub 2019 Oct 31. PMID: 31676222.

7. [Study Details | NCT05930704 | Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of SHR-2017 Injection in Postmenopausal Women | ClinicalTrials.gov](#)
8. [Study Details | NCT00078026 | An Investigation Into the Short Term Effect of APOMINE in Patients With Osteoporosis or Low Bone Mass | ClinicalTrials.gov](#)
9. [Study Details | NCT04435158 | A Multiple Dose Study to Evaluate the Effect of SHR-1222 Injection in Postmenopausal Osteoporosis Patients | ClinicalTrials.gov](#)
10. [Study Details | NCT00086619 | Parathyroid Hormone \(PTH\) for Osteoporosis in Postmenopausal Women | ClinicalTrials.gov](#)
11. [Study Details | NCT03183557 | Efficacy of Zoledronic Acid in Osteoporosis | ClinicalTrials.gov](#)
12. [Study Details | NCT06154187 | Study to Evaluate the Efficacy and Safety of PBK_L2201 in Postmenopausal Women With Osteoporosis | ClinicalTrials.gov](#)
13. [Study Details | NCT05067335 | A Study to Test the Efficacy, Safety and Tolerability of Romosozumab Treatment in Postmenopausal Chinese Women With Osteoporosis | ClinicalTrials.gov](#)
14. [Study Details | NCT05058976 | Romosozumab Use to Build Skeletal Integrity | ClinicalTrials.gov](#)
15. [Study Details | NCT05387200 | Phase IV Clinical Trial to Evaluate Efficacy and Safety of MASI BONE S \(Alendronate Sodium Trihydrate\) in Postmenopausal Women With Osteoporosis | ClinicalTrials.gov](#)